Readiness to Train

Prescriptive Analytics for Optimal Training Intensity

KU Leuven & OH Leuven — PhD Research Project
Project Overview & Problem Statement

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1. Core Objective

This project develops a **prescriptive analytics system** that estimates individual player **Readiness to Train** in professional football. Using **Causal Machine Learning** within a **Dynamic Treatment Regime (DTR)** framework, the system recommends an optimal daily training intensity for each player — delivered as a continuous score between 0 and 1. A score of 0 indicates the player should rest entirely; a score of 1 indicates the player can train at maximal intensity.

The system is fully **individualised**: each player receives a personalised score derived from their own physiological profile, training history, and evolving daily state. The problem is inherently a **Dynamic Treatment Regime with variable-length decision horizons** because (1) not every player is selected for every match — so some players skip a cycle entirely while others play — and (2) the time between consecutive matches varies (typically 5–8 days, but sometimes more during international breaks or cup fixtures). Each player's match-cycle length therefore differs both across players and across time.

■ ***The objective is NOT injury minimisation. A policy that minimises injury risk trivially prescribes zero load. The true objective is performance optimisation under biological constraints — where some non-zero injury incidence may be optimal because competitive performance requires physiological stress.***

2. Partnership & Data

The project is a collaboration between **KU Leuven** (methodology, analysis) and **OH Leuven** (professional football club, data provider). OH Leuven operates a systematic daily player monitoring programme covering GPS load metrics, subjective wellness questionnaires, medical status, and match performance. Four raw datasets are available:

Dataset	Rows	Columns	Players	Date Range	Granularity
Readiness_Data.xlsx	14,359	24	28	2024-07-01 → 2026-02-17 (597 days)	Daily (player-day)
Raw_Data.xlsx	9,968	39	84	2024-05-02 → 2026-03-01	Session-level (player-session)
Sessions.xlsx	1,206	9	—	2024-05-02 → 2026-03-01	Session-level (team-session)
Games.xlsx	403	9	24	2025-07-27 → 2026-02-28 (27 match dates)	Match-level (player-match)

The **processed dataset** merges all four sources. Readiness_Data is the base; Raw_Data provides detailed GPS/HR at session level (shifted +1 day so it appears as "yesterday" data); Games provides match performance data (also shifted +1 day). Player overlap: all 28 Readiness_Data players appear in Raw_Data; 23 of 28 appear in Games.

3. Why This Is a Causal Problem

Standard predictive modelling asks *"given the data, what will happen?"*. This project asks *"given a player's full history up to today, what training intensity today best contributes to future match-day performance?"*. This requires causal reasoning for three reasons:

1. **Time-Varying Confounding Affected by Prior Treatment.** Covariates at time t (fatigue, readiness, ACWR) are simultaneously *consequences* of treatment at t–1 and *causes* of treatment at t. Classical regression cannot handle this correctly — this is precisely the setting for which G-methods were developed.

2. **Treatment-Confounder Feedback.** Coaching decisions are observational. Hard sessions are prescribed when players appear fresh; recovery is prescribed when fatigue is high. The observed training-performance association conflates physiological response with coach selection behaviour.

3. **Sequential Decision-Making.** The treatment is a repeated, time-varying intervention. The question is not just "what training today?" but "what *sequence* of training intensities over the cycle maximises match-day performance?" — a DTR problem.

4. Causal Framework Variables

Symbol	Role	Variables
L(t) (Covariates)	Player state at time t	Wellness z-scores (fatigue, readiness, soreness, sleep, stress, mood), composite scores (Physical State, Mental State, Overall Wellbeing), ACWR metrics, Days Since Game, Days Until Match, medical availability, club attendance, raw GPS/HR (yesterday), match performance (yesterday — only filled day after match)
A(t) (Treatment)	Daily training intensity	Continuous score in [0,1], derived as tanh(mean of Total Distance %, High-Speed Distance %, Decelerations %, Sprints %) normalised against individual match benchmarks. GPS % columns are capped at 250% to remove extreme outliers. Free/rest days → 0.
Y (Outcome)	Match-day performance	High-intensity distance per Ball-In-Play minute (m/BIP). Continuous, to be maximised. Only filled in Games dataset on match-day rows (shifted +1 day in processed data).

5. Temporal Semantics (Critical)

Each row in the dataset represents **one player-day** (date t). Variables within a single row have different temporal positions. Understanding this is essential for avoiding data leakage and correctly specifying causal models.

Temporal position	Variables	Available at prediction?
Before day t (t–1 data)	Activity Type Yesterday, GPS % (×5), Training Intensity Yesterday, ACWR (×4), RPE Yesterday, Comment Yesterday, Total Minutes / Distance / HS Distance / HR / HR Exertion Yesterday (from Raw_Data), Match High Intensity / HIT Efforts / Minutes Played Yesterday (from Games, day after match only)	Yes — fully known

Temporal position	Variables	Available at prediction?
Morning of day t (assessment)	Wellness z-scores (fatigue, readiness, soreness, sleep quality, stress, mood), Physical State, Mental State, Overall Wellbeing, Status, Status Decrease, Days Since Game, Days Until Match, Match Day, Medical Availability, Club Attendance, Position	Yes — measured before activity decisions
Post-assessment (same day t)	Activity Type Today, Selected	No — assigned AFTER morning assessment. Only valid as treatment A(t) in causal analyses, never as a predictor in standard ML.

Critical rules:

Days Since Game is NEVER 0. It counts days since the last completed OHL first-team game. On match day the game has not yet occurred at the morning assessment, so the count refers to the previous match. Minimum value: 1.

Days Until Match CAN be 0 — on match day itself for selected players. Only OHL first-team matches are counted.

Match Day is schedule information — although derived from Activity Type Today in preprocessing, it represents the team fixture list (known in advance) and is safe to use as a predictor.

Wellness z-scores are individualised — each player's z-scores are relative to their own 28-day rolling baseline, not the team average.

Lag features are player-grouped — shifted values (e.g. Fatigue (z) at t–1) are created within each player's own time series to prevent cross-player data leakage.

6. Sports Science Foundation

Supercompensation

Training stress produces fatigue, which, during recovery, is followed by adaptation and an increase in performance capacity above baseline. This cycle — stress → fatigue → recovery → supercompensation — underpins all periodisation theory. The DTR framework seeks the sequence of training intensities that maximises supercompensation at match day, neither under-loading (insufficient stimulus) nor over-loading (accumulated fatigue with insufficient recovery).

Acute:Chronic Workload Ratio (ACWR)

ACWR is the ratio of a player's recent acute load (7-day exponential moving average) to their chronic load (42-day EMA). It captures where a player sits on the stress-recovery-adaptation curve. The dataset provides ACWR for four GPS metrics: Total Distance, High-Speed Distance, Decelerations (>3 m/s²), and Sprints.

ACWR Zone	Interpretation
< 0.8	Under-trained — insufficient stimulus for adaptation
0.8 – 1.3	Sweet spot — optimal fitness-fatigue balance
1.3 – 1.5	Caution — elevated fatigue accumulation
> 1.5	Danger zone — flagged as Any ACWR Danger = 1; injury risk sharply elevated (Gabbett, 2016)

Individual Profiling

GPS metrics are expressed as a percentage of each player's own match-day benchmarks (e.g. Total Distance % = session distance / player's average match distance). This normalises for positional differences and individual fitness levels, making comparisons across players meaningful. Values are capped at 250% to remove extreme outliers from low-benchmark or early-season matches. Wellness z-scores use a 28-day rolling window per player. The training intensity score A(t) is the tanh-compressed mean of these GPS percentages, soft-capped in [0, 1).

Match Cycle Structure

The typical weekly structure consists of 5–6 training days followed by a match day. This creates the **match cycle** structure central to the DTR formulation. Each cycle begins the day after a match and ends on the next match day. Cycle boundaries are **player-specific**: days where the team plays but a given player is not selected simply extend that player's current cycle.

7. Key Challenges

Small N, Large T

Only 28 players are monitored in Readiness_Data, but each player has up to 597 days of observations. The signal is likely dominated by **within-player dynamics**. Models must be parsimonious or leverage partial pooling to avoid overfitting across players. Methods must balance individual-level tailoring with the limited between-player sample size.

Time-Varying Confounding Affected by Prior Treatment

Covariates L(t) at time t are simultaneously consequences of treatment at t–1 and causes of treatment at t. This is the defining feature of the problem and the reason standard regression fails. The appropriate causal identification framework is the theory of G-methods (Robins, 1986): G-computation, Marginal Structural Models with IPTW, or G-Estimation / Structural Nested Mean Models.

Observational Data with Coach Selection Bias

Training loads are prescribed by coaching staff based on observed player state. Players who appear fresh receive harder sessions; fatigued players receive recovery sessions. This creates a systematic association between baseline health and training dose that confounds any naive regression of load on performance.

Sparse Match-Day Outcomes

The outcome Y (match performance) is observed only on match days. With a 1-week cycle, each player contributes approximately one outcome observation per 6 rows of covariate/treatment data. The Games dataset covers 24 players over 27 match dates (403 player-match rows). Mean minutes played: 67.4 min/match; mean high-intensity distance per BIP: 18.75 m/min.

8. Key Research Questions

RQ1. How can we accurately estimate the effect of training intensity sequences in a "Small N, Large T" observational environment (28 players, up to 597 days) where standard statistical assumptions are violated by time-varying confounding?

RQ2. How do we handle **time-varying confounding affected by prior treatment** in observational football data, where coaching decisions create systematic confounding between player state and prescribed load?

RQ3. Can a causal DTR model identify the optimal individualised training intensity sequence — one per day in the pre-match cycle — that maximises expected match-day performance for each player?